

TEXAS GULF COAST REGIONAL, TX, USA

WMO: 722527

Lat: 29.110N

Lon: 95.462W

Elev: 25

StdP: 14.68

Time Zone: -6.00 (NAC)

Period: 98-19

WBAN: 12976

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	31.1	34.8	19.4	14.6	41.9	24.2	18.4	45.3	24.8	66.2	22.9	63.9	4.8	10	0.411

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	17.4	94.9	77.9	93.0	78.0	91.4	77.9	81.1	88.6	80.4	87.8	79.8	87.2	9.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.3	152.3	83.5	78.8	149.7	83.5	77.9	144.9	83.3	44.6	88.5	43.8	87.8	43.2	86.8	86.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.5	19.3	17.6	DB	25.7	98.6	3.4	2.9	23.3	100.7	21.3	102.5	19.4	104.1	17.0	106.2	
			WB	23.8	82.5	3.4	1.4	21.3	83.5	19.4	84.3	17.5	85.0	15.1	86.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	70.1	54.3	58.3	63.6	69.8	76.4	81.5	82.7	83.4	79.7	71.7	63.0	56.6
	DBStd	12.36	9.61	9.56	7.95	6.33	4.87	2.64	2.43	2.56	4.14	7.13	9.00	9.53
	HDD50	154	64	32	6	0	0	0	0	0	0	10	42	
	HDD65	1200	352	226	124	30	3	0	0	0	26	145	294	
	CDD50	7502	197	263	428	593	817	945	1015	1034	892	673	398	247
	CDD65	3073	20	37	81	173	355	495	550	569	442	234	83	34
	CDH74	29539	53	97	273	986	3082	5402	6192	6500	4259	2065	495	135
	CDH80	11601	1	3	18	128	861	2276	2815	3091	1743	598	64	3
Wind	WSAvg	7.5	8.3	8.9	9.0	9.4	9.1	7.2	5.8	5.4	5.6	6.6	7.3	8.0
Precipitation	PrecAvg	49.3	4.1	2.8	3.2	3.8	3.9	4.6	3.8	4.0	5.6	4.7	4.3	3.9
	PrecMax	70.6	11.5	8.6	8.8	9.1	10.4	11.9	12.7	11.9	14.8	13.9	14.4	9.7
	PrecMin	22.8	0.0	0.4	0.0	0.4	0.0	0.6	0.0	0.7	1.1	0.0	0.9	0.7
	PrecStd	11.9	2.8	2.1	2.1	2.5	2.8	2.7	3.1	3.1	3.7	3.7	3.1	2.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	78.8	79.7	82.1	85.9	90.5	95.9	96.3	98.2	97.2	90.1	84.5	80.4
		MCWB	68.1	68.9	69.2	71.0	75.2	76.2	77.8	77.9	76.4	75.9	73.4	71.2
	2%	DB	75.3	77.0	79.5	82.9	88.2	91.9	93.7	95.3	92.7	88.0	81.8	77.7
		MCWB	66.3	68.4	69.6	71.9	75.5	77.1	78.2	78.5	76.6	75.9	72.6	70.4
	5%	DB	72.8	74.4	77.1	81.3	86.4	90.5	92.1	93.2	90.6	85.8	79.4	75.1
		MCWB	66.5	67.5	68.6	71.7	75.1	77.2	78.3	78.6	76.9	74.4	70.8	69.5
	10%	DB	70.3	72.3	74.8	79.4	84.5	89.5	90.7	91.5	88.6	83.6	76.8	72.6
		MCWB	65.5	67.0	67.9	70.5	74.4	77.3	78.4	78.5	76.6	73.3	69.7	68.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.6	72.4	73.3	77.0	80.5	81.3	81.8	82.3	81.3	80.1	76.4	74.3
		MCDB	75.3	76.1	78.4	82.4	84.5	87.9	89.6	89.7	89.1	83.9	80.4	77.2
	2%	WB	70.1	70.9	72.0	75.0	78.4	80.2	80.8	81.2	80.2	78.4	75.0	72.5
		MCDB	73.4	74.3	76.2	79.9	83.5	86.7	88.5	89.0	87.3	83.6	79.2	75.8
	5%	WB	68.6	69.7	71.0	73.9	77.3	79.4	80.2	80.4	79.3	77.3	73.3	71.2
		MCDB	71.7	72.9	74.7	78.6	83.2	85.9	87.8	88.4	86.1	83.0	77.4	74.2
	10%	WB	66.5	68.4	70.0	73.0	76.3	78.5	79.5	79.7	78.4	76.1	71.8	69.1
		MCDB	69.7	71.4	73.5	77.3	82.1	85.4	87.1	87.8	85.0	81.6	75.7	72.1
Mean Daily Temperature Range	5% DB	MDBR	19.2	17.9	17.9	17.3	16.1	15.8	16.3	17.4	17.1	20.2	20.0	19.2
		MCDBR	19.5	17.4	16.5	16.7	16.9	17.6	18.3	19.4	19.5	19.3	18.7	19.6
		MCWBR	13.2	10.9	8.7	8.4	6.8	6.2	6.2	6.3	7.1	8.8	10.5	13.4
	5% WB	MCDBR	16.4	14.7	13.1	13.5	13.1	14.5	16.0	16.9	15.6	16.0	15.6	17.5
		MCWBR	13.4	10.8	8.1	7.7	6.1	6.3	6.3	6.3	6.7	8.9	10.7	13.7
Clear-Sky Solar Irradiance	taub	0.350	0.364	0.379	0.438	0.482	0.464	0.492	0.478	0.454	0.384	0.377	0.358	
	taud	2.497	2.468	2.412	2.248	2.158	2.267	2.207	2.248	2.283	2.469	2.435	2.480	
	Ebn at Noon	281	287	288	273	260	263	256	259	262	276	268	272	
	Edh at Noon	29	32	36	44	49	43	46	44	41	32	31	29	
All-Sky Solar Radiation	RadAvg	918	1051	1433	1711	1926	2017	1940	1863	1595	1403	1056	820	
	RadStd	94	173	125	134	131	147	184	138	164	154	108	99	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (32 neighbors)	+0.58	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+203	+178

Nomenclature: See separate page